

(12) **United States Patent**  
**Chang et al.**

(10) **Patent No.:** **US 12,412,783 B2**  
(45) **Date of Patent:** **\*Sep. 9, 2025**

(54) **SEMICONDUCTOR DEVICE**

(71) Applicant: **Parabellum Strategic Opportunities Fund LLC**, Wilmington, DE (US)

(72) Inventors: **Chia-Hao Chang**, Hsinchu (TW);  
**Jia-Chuan You**, Taoyuan (TW);  
**Yu-Ming Lin**, Hsinchu (TW);  
**Chih-Hao Wang**, Hsinchu (TW);  
**Wai-Yi Lien**, Hsinchu (TW)

(73) Assignee: **Parabellum Strategic Opportunities Fund LLC**, Austin, TX (US)

(\*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/662,544**

(22) Filed: **May 13, 2024**

(65) **Prior Publication Data**

US 2024/0379431 A1 Nov. 14, 2024

**Related U.S. Application Data**

(60) Continuation of application No. 17/874,170, filed on Jul. 26, 2022, now Pat. No. 12,009,257, which is a continuation of application No. 16/871,983, filed on May 11, 2020, now Pat. No. 11,404,315, which is a  
(Continued)

(51) **Int. Cl.**

**H01L 23/535** (2006.01)  
**H01L 21/768** (2006.01)  
**H01L 23/528** (2006.01)  
**H10D 30/69** (2025.01)  
**H10D 62/00** (2025.01)

**H10D 64/01** (2025.01)  
**H10D 62/13** (2025.01)  
**H10D 62/822** (2025.01)  
(52) **U.S. Cl.**  
CPC .. **H01L 21/76895** (2013.01); **H01L 21/76805** (2013.01); **H01L 21/76816** (2013.01); **H01L 21/76831** (2013.01); **H01L 23/528** (2013.01); **H01L 23/535** (2013.01); **H10D 30/797** (2025.01); **H10D 62/021** (2025.01); **H10D 64/017** (2025.01); **H10D 62/151** (2025.01); **H10D 62/822** (2025.01)

(58) **Field of Classification Search**  
CPC ..... H01L 23/528; H01L 23/535; H01L 29/66545; H01L 29/66636; H01L 29/7848; H01L 29/0847; H01L 29/165  
See application file for complete search history.

(56) **References Cited**

**U.S. PATENT DOCUMENTS**

9,614,047 B2 4/2017 Horak et al.  
9,666,533 B1 5/2017 Basker et al.  
9,748,349 B2 8/2017 Lu et al.  
(Continued)

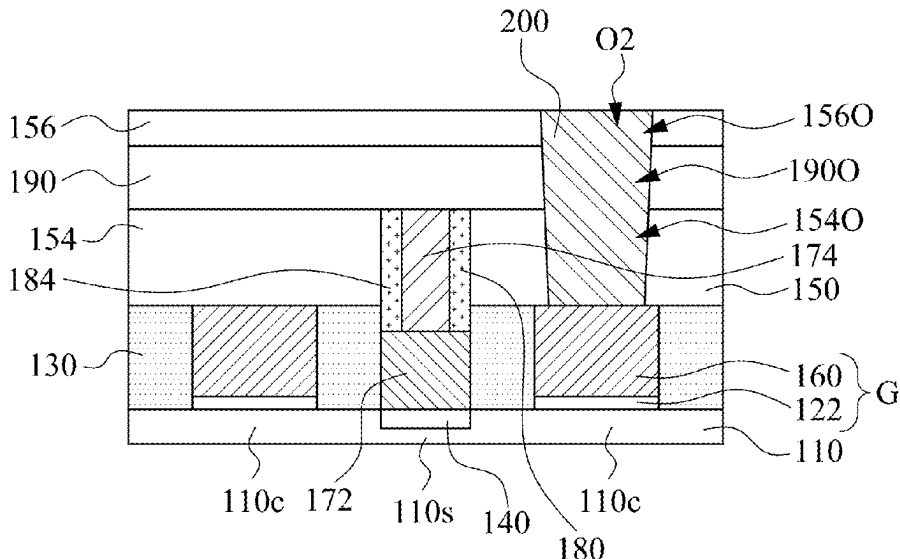
*Primary Examiner* — Samuel A Gebremariam

(74) *Attorney, Agent, or Firm* — Almanac IP Advisors LLP

(57) **ABSTRACT**

A semiconductor device includes a semiconductor substrate, a gate electrode, a source/drain contact, a conductive structure, and a dielectric liner. The semiconductor substrate has a channel region and a source/drain region. The gate electrode is over the channel region. The source/drain contact is over the source/drain region. The conductive structure is over a top surface of the source/drain contact. The dielectric liner surrounds the conductive structure and is over the top surface of the source/drain contact.

**12 Claims, 8 Drawing Sheets**



**Related U.S. Application Data**

division of application No. 15/719,395, filed on Sep.  
28, 2017, now Pat. No. 10,651,085.

(56)

**References Cited**

## U.S. PATENT DOCUMENTS

|              |      |         |                          |                          |
|--------------|------|---------|--------------------------|--------------------------|
| 10,002,789   | B2   | 6/2018  | Adusumilli et al.        |                          |
| 2014/0134836 | A1   | 5/2014  | Pranatharthiharan et al. |                          |
| 2016/0064518 | A1 * | 3/2016  | Liu .....                | H01L 21/76895<br>438/586 |
| 2016/0172300 | A1   | 6/2016  | Lu et al.                |                          |
| 2016/0233164 | A1   | 8/2016  | Choi et al.              |                          |
| 2016/0343825 | A1 * | 11/2016 | Bae .....                | H10D 62/021              |
| 2016/0379925 | A1   | 12/2016 | Ok et al.                |                          |
| 2017/0053997 | A1   | 2/2017  | Cheng et al.             |                          |

\* cited by examiner

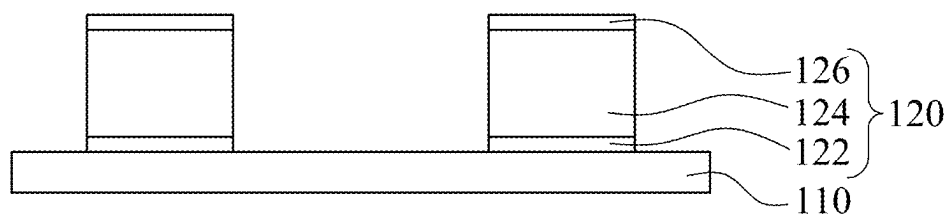


Fig. 1

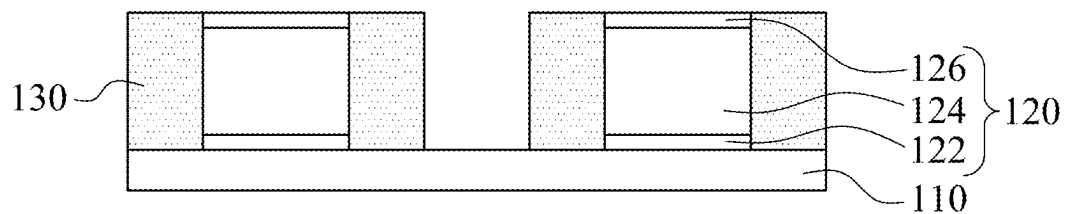


Fig. 2

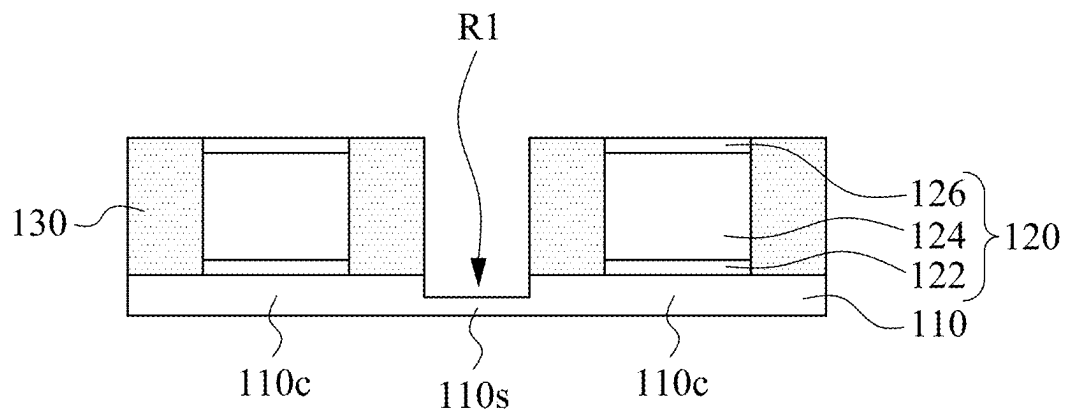


Fig. 3

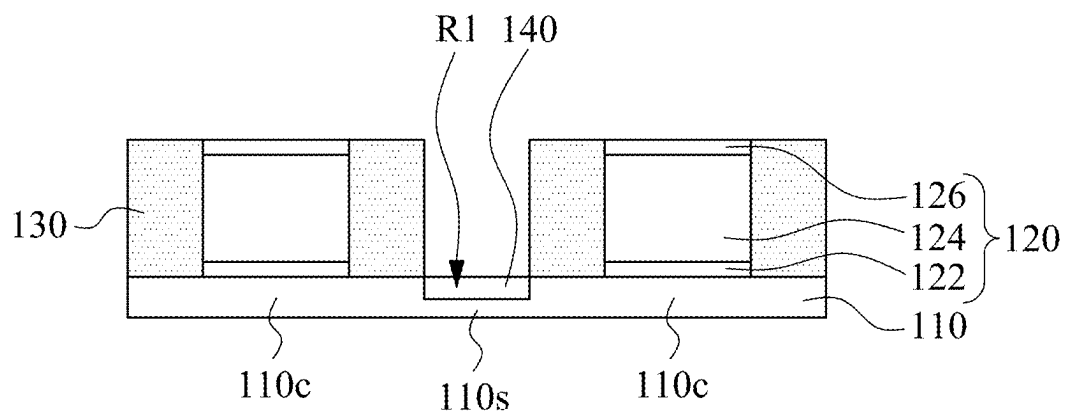


Fig. 4

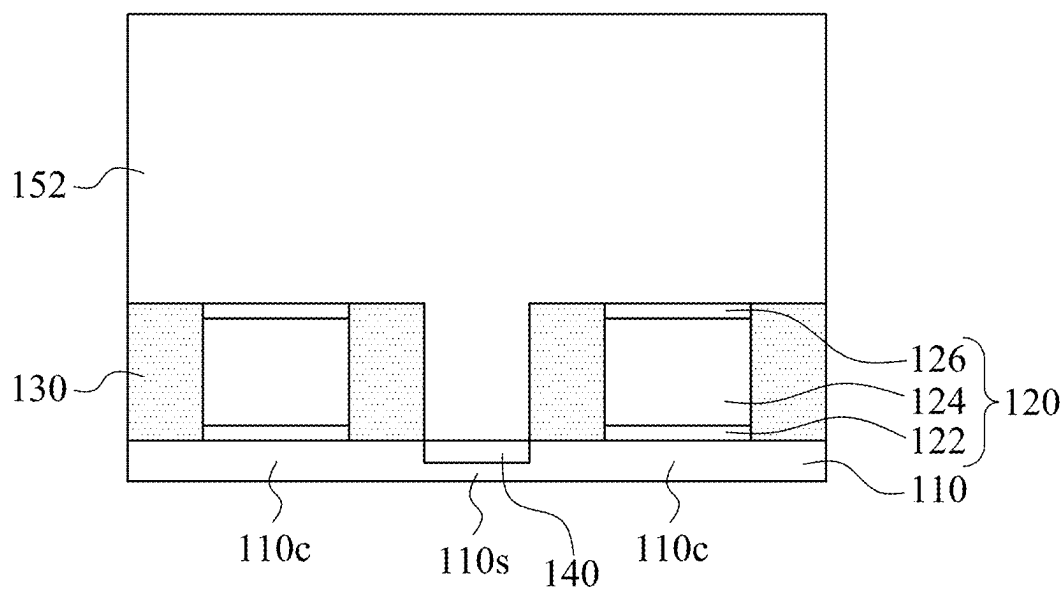


Fig. 5

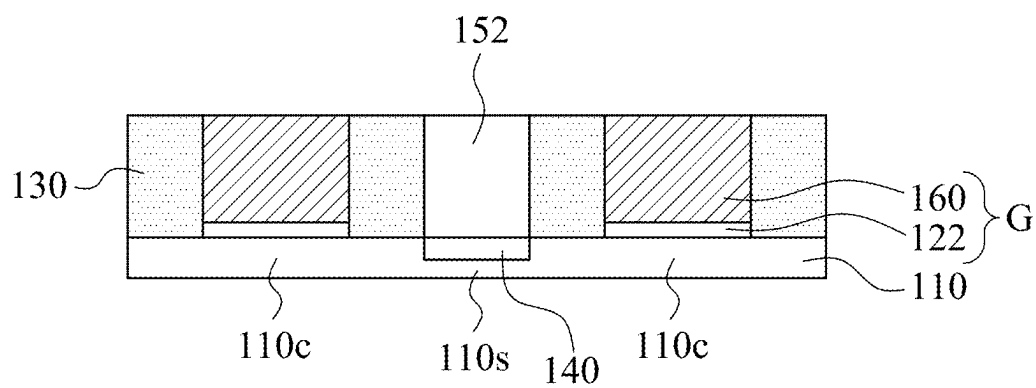


Fig. 6

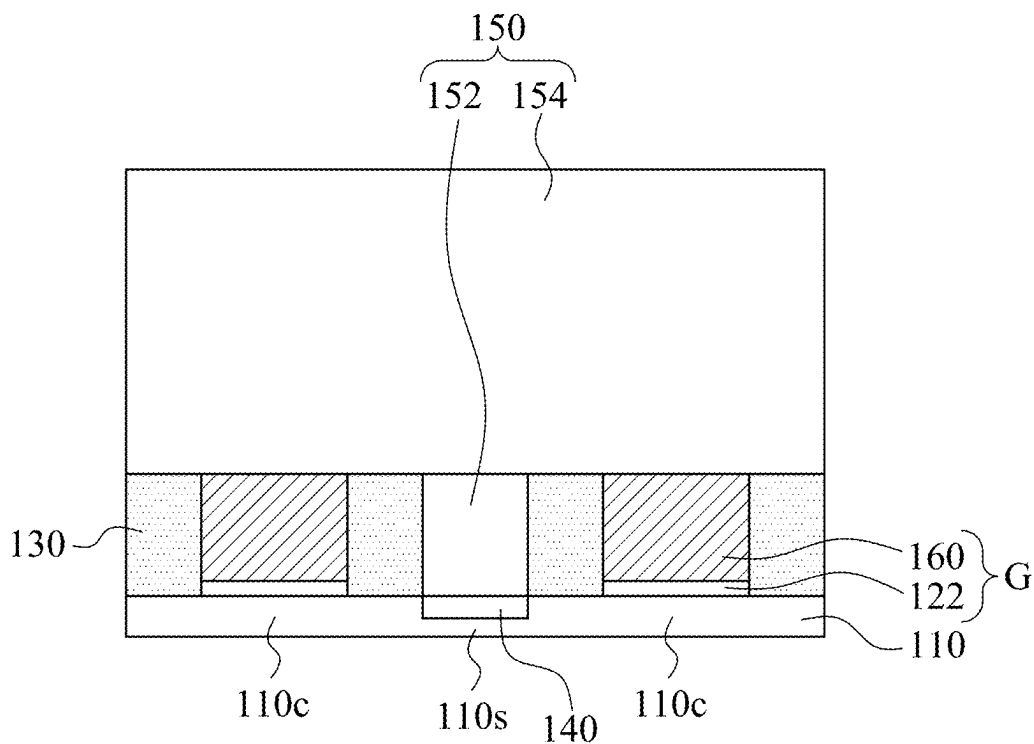


Fig. 7

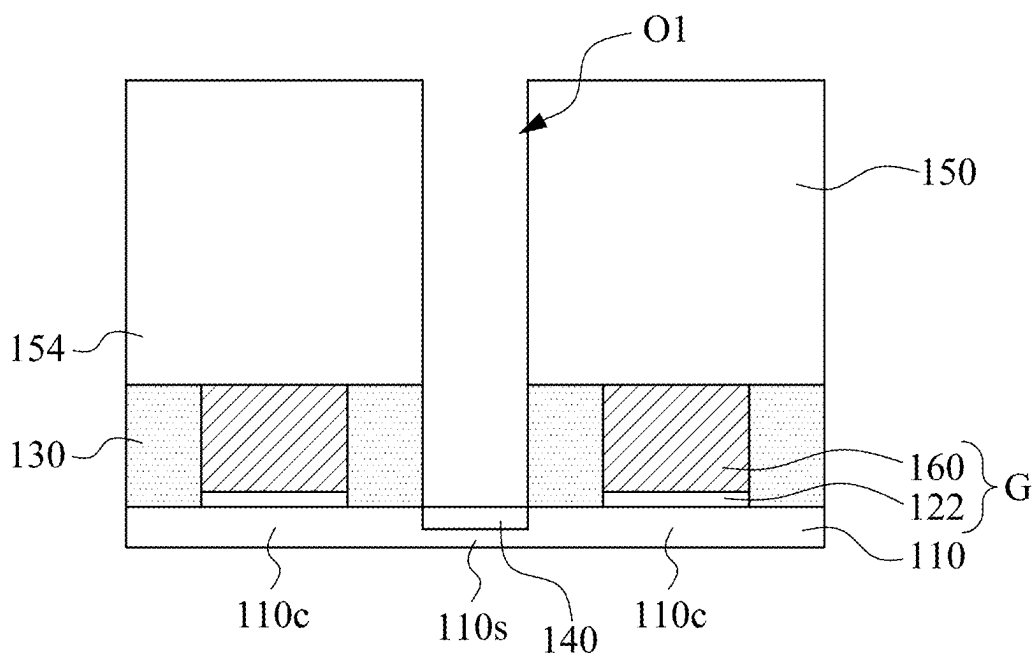


Fig. 8

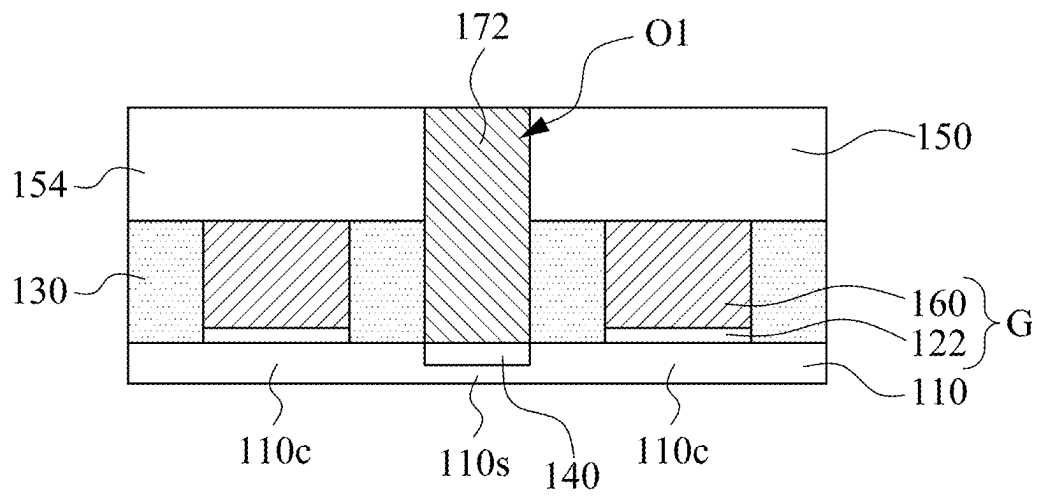


Fig. 9

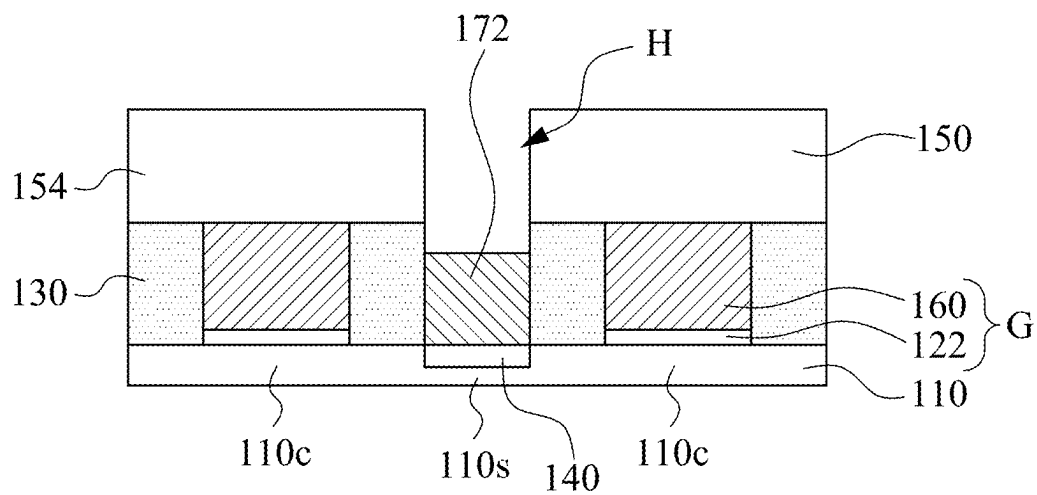


Fig. 10

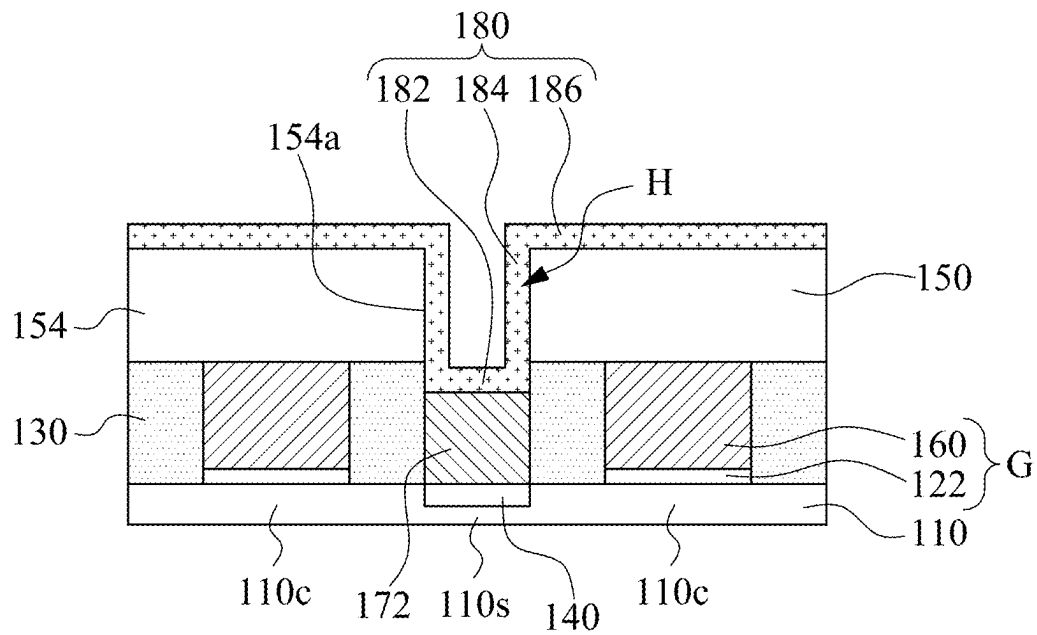


Fig. 11

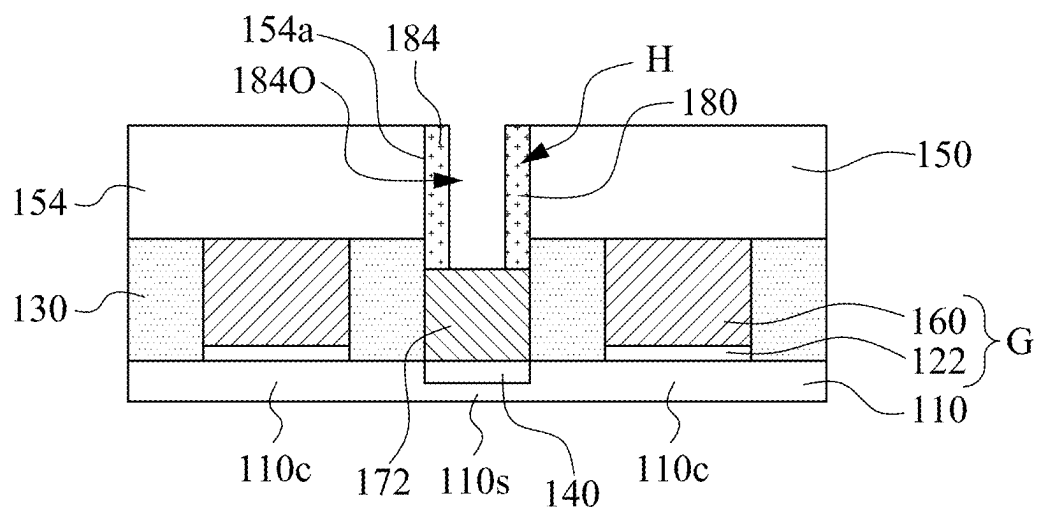


Fig. 12



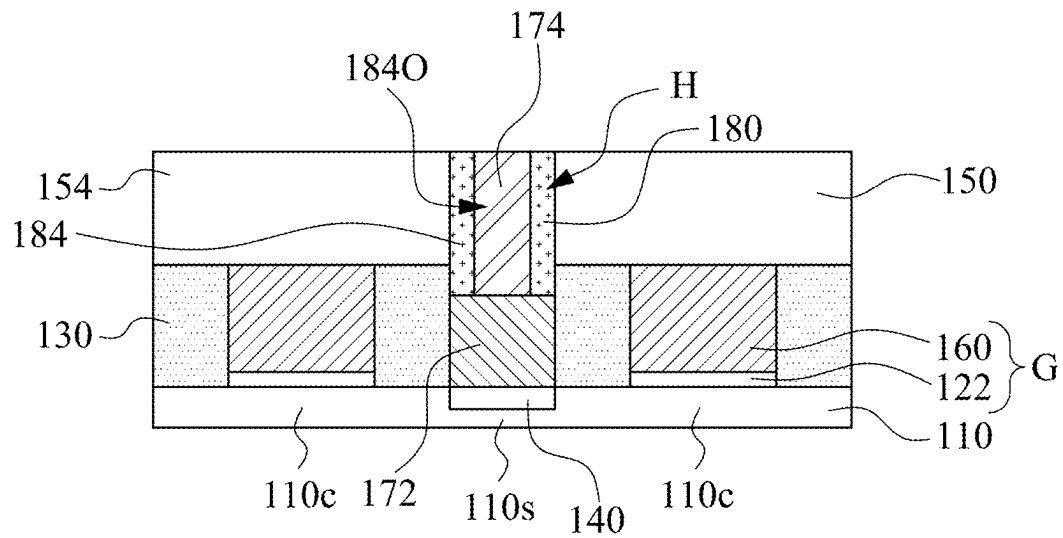


Fig. 13

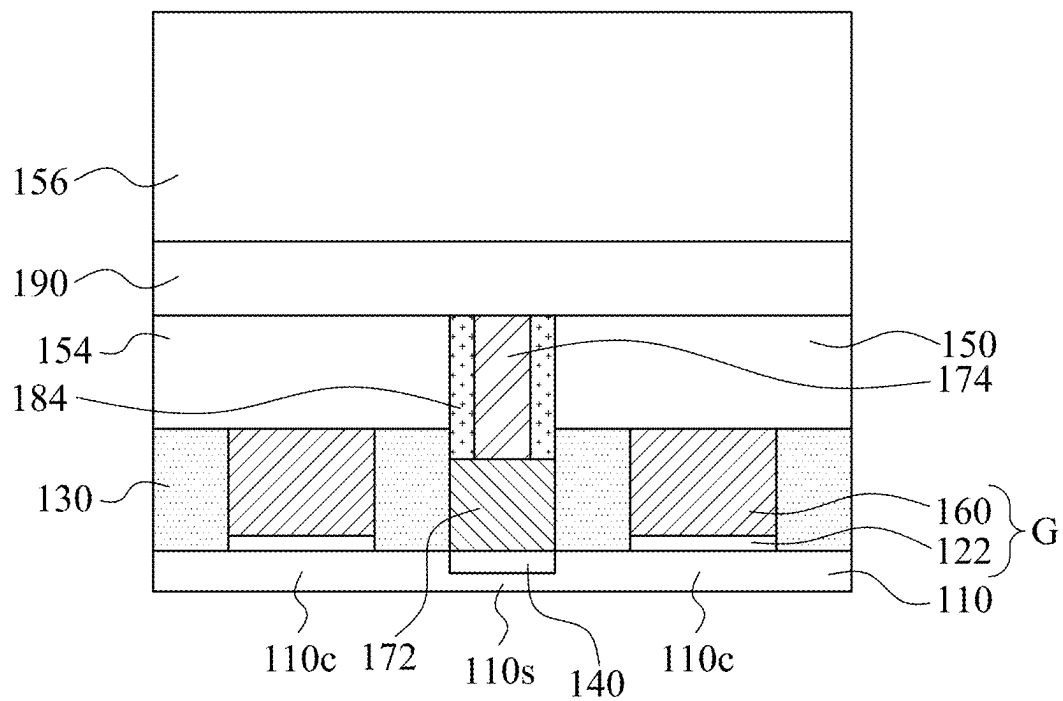


Fig. 14

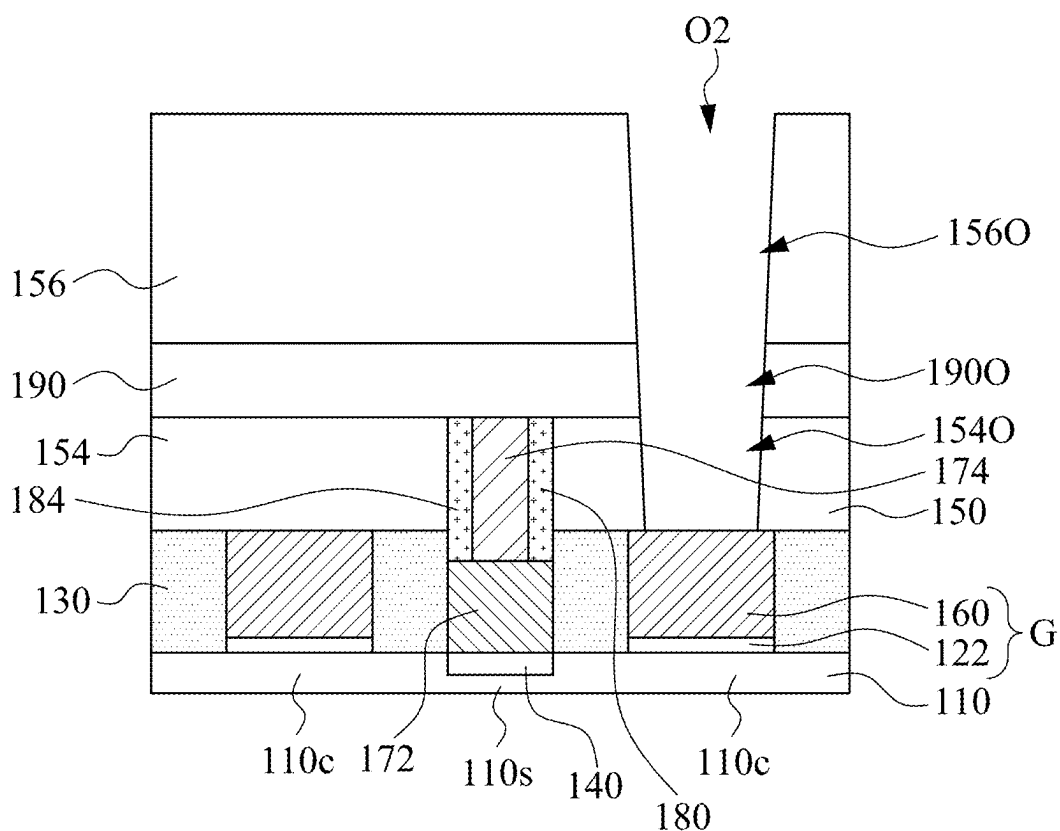


Fig. 15

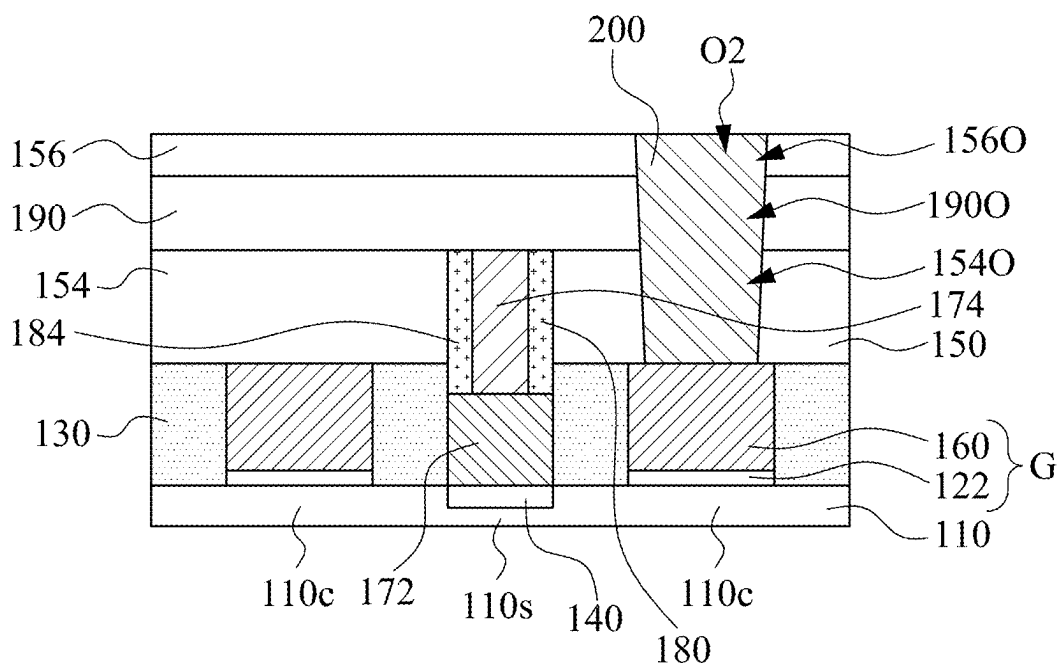


Fig. 16

## SEMICONDUCTOR DEVICE

## PRIORITY CLAIM AND CROSS-REFERENCE

This application is a continuation application of the application Ser. No. 17/874,170, filed on Jul. 26, 2022, which is a continuation application of the application Ser. No. 16/871,983, filed on May 11, 2020, now U.S. Pat. No. 11,404,315, issued Aug. 2, 2022, which is a division application of the U.S. patent application Ser. No. 15/719,395, filed on Sep. 28, 2017, now U.S. Pat. No. 10,651,085, issued May 12, 2020, the entirety of which is incorporated by reference herein in their entireties.

## BACKGROUND

The semiconductor integrated circuit (IC) industry has experienced rapid growth. Technological advances in IC materials and design have produced generations of ICs. Each generation has smaller and more complex circuits than the previous generation. However, these advances have increased the complexity of processing and manufacturing ICs.

In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometric size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. However, since the feature sizes continue to decrease, fabrication processes continue to become more difficult to perform. Therefore, it is a challenge to form reliable semiconductor devices with smaller and smaller sizes.

## BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 to FIG. 16 are cross-sectional views of a method for manufacturing a semiconductor device at various stages in accordance with some embodiments of the present disclosure.

## DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and

clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Fins in FinFETs may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in some embodiments, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins.

FIG. 1 to FIG. 16 are cross-sectional views of a method for manufacturing a semiconductor device at various stages in accordance with some embodiments of the present disclosure.

Reference is made to FIG. 1. A semiconductor substrate 110 is provided. In some embodiments, the semiconductor substrate 110 may be a semiconductor fin protruding from a substrate. Gate structures 120 are formed on the semiconductor substrate 110. In some embodiments, the semiconductor substrate 110 includes silicon. Alternatively, the semiconductor substrate 110 may include germanium, silicon germanium, gallium arsenide or other appropriate semiconductor materials. Also alternatively, the semiconductor substrate 110 may include an epitaxial layer. Further, the semiconductor substrate 110 may be strained for performance enhancement. For example, the epitaxial layer may include a semiconductor material different from that of the bulk semiconductor, such as a layer of silicon germanium overlying bulk silicon or a layer of silicon overlying bulk silicon germanium. Furthermore, the semiconductor substrate 110 may include a semiconductor-on-insulator (SOI) structure. Also alternatively, the semiconductor substrate 110 may include a buried dielectric layer, such as a buried oxide (BOX) layer, such as that formed by separation by implantation of oxygen (SIMOX) technology, wafer bonding, or other appropriate method. In some embodiments, the semiconductor substrate 110 includes silicon. The semiconductor substrate 110 may be a semiconductor fin protruding from a substrate.

In some embodiments, the gate structures 120 can serve as a dummy gate structure and at least portions thereof will be replaced with a replacement gate structure using a “gate-last” or replacement-gate process. For example, the dummy gate structures 120 may be replaced later by metal gate electrodes (MG) after high temperature thermal processes, such as thermal annealing for source/drain activation during the sources/drains formation. In other embodiments, the gate structures 120 are active gates and are formed in a “gate-first process” and will not be replaced.

The gate structures **120** can be formed by deposition and patterning. In some embodiments, the gate structures **120** each include a gate dielectric layer **122**, a dummy gate electrode **124** with an overlying mask layer **126**. The gate dielectric layer **122** is blanket deposited on the semiconductor substrate **110** by a suitable technique, such as thermal oxidation, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), sputtering, other suitable processes, or combinations thereof. In some embodiments, the gate dielectric layer **122** may include, for example, a high-k dielectric material such as metal oxides, metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates, oxynitrides of metals, metal aluminates, zirconium silicate, zirconium aluminate, or combinations thereof. In some

embodiments, the gate dielectric layer **122** may have a multilayer structure such as one layer of silicon oxide (e.g., interfacial layer) and another layer of high-k material. The dummy gate electrode **124** is deposited on the gate dielectric layer **122** by a suitable technique, such as thermal oxidation, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), sputtering, other suitable processes, or combinations thereof. In some embodiments, the dummy gate electrode **124** may include polycrystalline-silicon (poly-Si), poly-crystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metallic oxides, or metals. In some embodiments, the dummy gate electrode **124** may include a metal-containing material such as TiN, TaN, TaC, Co, Ru, Al, combinations thereof, or multi-layers thereof.

The mask layer **126**, such as photoresists, hard masks, combinations thereof, or multi-layers thereof, may be formed over the dummy gate electrode **124**. Then, the mask layer **126** is patterned by a lithography process and an etching process, thereby forming openings in the mask layer **126**, exposing the underlying dummy gate electrode layer within the openings. The lithography process may include photoresist (or resist) coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, developing the photoresist, rinsing, drying (e.g., hard baking), other suitable processes, and/or combinations thereof. The etching process includes dry etching, wet etching, and/or other etching methods (e.g., reactive ion etching). Another etching process is applied to the dummy gate electrode layer and the gate dielectric layer through the openings of the mask layer **126**, thereby forming the dummy gate structures **120** straddling portions of the semiconductor substrate **110** as shown in FIG. 1, if the semiconductor substrate **110** is a semiconductor fin.

Reference is made to FIG. 2. Gate spacers **130** are formed on sidewalls of the gate structures **120**. In some embodiments, the gate spacers **130** may include silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, silicon carbonitride, silicon oxycarbonitride, silicon oxycarbide or other suitable material. The gate spacers **130** may include a single layer or multilayer structure. To form the gate spacers **130**, a blanket layer may be formed on the semiconductor substrate **110** by CVD, PVD, ALD, or other suitable technique. In some embodiments, the gate spacers **130** are used to offset subsequently formed doped regions, such as source/drain regions. The gate spacers **130** may further be used for designing or modifying the source/drain region (junction) profile.

Reference is made to FIG. 3. At least one portion of the semiconductor substrate **110** uncovered by the gate structure **120** and the gate spacers **130** is removed (or recessed) to form at least one source/drain recess R1 in the semiconduc-

tor substrate **110**. A remaining portion of the semiconductor substrate **110** has at least one source/drain portion **110s** and channel portions **110c**. The channel portions **110c** underlie the gate structures **120**, and the source/drain portion **110s** is not covered by the dummy gate structures **120** and the gate spacers **130**.

Reference is made to FIG. 4. At least one epitaxy feature **140** (also referred to as a source/drain region) is formed within the recess R1 and on the source/drain portions **110s** of the semiconductor substrate **110**. In some embodiments, the epitaxy feature **140** may be formed using one or more epitaxy or epitaxial (epi) processes, such that Si features, SiGe features, SiP features, and/or other suitable features can be formed in a crystalline state on the source/drain portion **110s** of the semiconductor substrate **110**. In some embodiments, the lattice constant of the epitaxy feature **140** is different from the lattice constant of the semiconductor substrate **110**, so that the channel portions **110c** of the semiconductor substrate **110** can be strained or stressed by the epitaxy feature **140** to improve carrier mobility of the semiconductor device and enhance the device performance. The epitaxy processes include CVD deposition techniques (e.g., vapor-phase epitaxy (VPE) and/or ultra-high vacuum CVD (UHV-CVD)), molecular beam epitaxy, and/or other suitable processes. The epitaxy process may use gaseous and/or liquid precursors, which interact with the composition of the semiconductor substrate **110**.

Reference is made to FIG. 5. A first inter-layer dielectric (ILD) layer **152** is formed over the semiconductor substrate **110**. The first ILD layer **152** includes silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, low-dielectric constant dielectric material or a combination thereof. The first ILD layer **152** includes a single layer or multiple layers. The first ILD layer **152** is formed by a suitable technique, such as CVD. Subsequently, a chemical mechanical planarization (CMP) process is applied to remove excessive first ILD layer **152** until the mask layer **126** is removed, and the resulting structure is shown in FIG. 6. In other words, the first ILD layer **152** is removed to expose a top surface of the dummy gate electrodes **124** for a subsequent gate replacement process.

Thereafter, the gate replacement process is performed to replace the dummy gate structures **120** with gate electrodes **160** (or gate conductors) respectively. More particularly, dummy gate electrodes **124** of the dummy gate structures **120** (as shown in FIG. 5) are removed to form gate trenches with the gate spacers **130** as their sidewalls, and the gate electrodes **160** are formed in the gate trenches respectively. In some embodiments, the gate dielectric layers **122** are removed as well. The dummy gate electrodes **124** may be removed by dry etch, wet etch, or a combination of dry and wet etch. For example, a wet etch process may include exposure to a hydroxide containing solution (e.g., ammonium hydroxide), deionized water, and/or other suitable etchant solutions.

In some embodiments, one of the gate electrodes **160** and one of the underlying gate dielectric layers **122** can be collectively referred to as a gate stack G, as shown in FIG. 6. The gate stacks G straddle the semiconductor substrate **110** and extend along the gate spacers **130**, if the semiconductor substrate **110** is formed as a semiconductor fin. In some embodiments, one of the gate electrodes **160** includes a work function conductor and a filling conductor in a recess of the work function conductor. For example, the work function conductor of the gate electrodes **160** may include one or more n-type work function metals (N-metal), such as titanium aluminide (TiAl), titanium aluminium nitride

5

(TiAlN), carbo-nitride tantalum (TaCN), hafnium (Hf), zirconium (Zr), titanium (Ti), tantalum (Ta), aluminum (Al), metal carbides (e.g., hafnium carbide (HfC), zirconium carbide (ZrC), titanium carbide (TiC), aluminum carbide (AlC)), aluminides, and/or other suitable materials. Alternatively, the work function conductor may include one or more p-type work function metals (P-metal), such as titanium nitride (TiN), tungsten nitride (WN), tungsten (W), ruthenium (Ru), palladium (Pd), platinum (Pt), cobalt (Co), nickel (Ni), conductive metal oxides, and/or other suitable materials. In some embodiments, the filling conductor of the gate electrodes **160** may exemplarily include, but are not limited to, tungsten, aluminum, copper, nickel, cobalt, titanium, tantalum, titanium nitride, tantalum nitride, nickel silicide, cobalt silicide, TaC, TaSiN, TaCN, TiAl, TiAlN, or other suitable materials.

Reference is made to FIG. 7. A second ILD layer **154** is formed on the first ILD layer **152** and the gate electrode **160**. The first ILD layer **152** and the second ILD layer **154** can be collectively referred to as an ILD structure **150**. The second ILD layer **154** may include substantially the same materials as first ILD layer **152** in some embodiments. In some embodiments, the first and second ILD layers **152** and **154** have different dielectric materials. In some embodiments, the second ILD layer **154** includes silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, low-dielectric constant dielectric material or a combination thereof. The second ILD layer **154** includes a single layer or multiple layers. The second ILD layer **154** is formed by a suitable technique, such as CVD.

Reference is made to FIG. 8. A source/drain contact hole **O1** is formed in the ILD structure **150** to expose a portion of the epitaxy feature **140**. In some embodiments, the source/drain contact hole **O1** is etched in the second and first ILD layers **154** and **152** until reaching the epitaxy feature **140**. In some embodiments, the source/drain contact hole **O1** may be formed, for example, by patterning and etching the ILD structure **150** using photolithography techniques. For example, a layer of photoresist material (not shown) is deposited over the ILD structure **150**. The layer of photoresist material is irradiated (exposed) in accordance with a desired pattern (the source/drain contact hole **O1** in this case) and developed to remove portions of the photoresist material. The remaining photoresist material protects the underlying material from subsequent processing steps, such as etching. The etching process, such as a dry etching, wet etching, and/or plasma etching process, is performed to remove portions of the first ILD layer **152** and the second ILD layer **154**. In some embodiments, the etching process may be, but not limited to be, anisotropic etching. In some other embodiments, the source/drain contact hole **O1** is etched in the first and second ILD layers **152** and **154** until reaching the epitaxy feature **140**.

Next, a source/drain contact **172** is formed in the source/drain contact hole **O1** and is in contact with the epitaxy feature **140**, and a CMP process is carried out to planarize the source/drain contact **172** with the second ILD layer **154**. The resulting structure is shown in FIG. 9. Exemplary formation method of the source/drain contact **172** may include depositing metal or other suitable conductive materials in the source/drain contact hole **O1** using a deposition process, such as a CVD process.

FIG. 10 illustrates etching back the source/drain contact **172**. Herein, the etching back process is selective to the source/drain contact **172**. To be specific, the second ILD layer **154** has higher etch resistance to the etching back than that of the source/drain contact **172**, such that a top surface

6

of the etched back source/drain contact **172** is lower than that of the second ILD layer **154** after the etching back. As a result of the etching back process, a hole **H** is formed in the second ILD layer **154** and over the source/drain contact **172**. In some embodiments, the source/drain contact **172** is etched back such that the top surface of the resulting source/drain contact **172** is lower than the top surfaces of the gate electrodes **160**. For example, the source/drain contact **172** is selectively etched back until reaching a position lower than top surfaces of gate electrodes **160** proximate the epitaxy feature **140**. In some embodiments, after the source/drain contact **172** is etched back, upper portions of sidewalls of the gate spacers **130** are exposed by the hole **H**.

Reference is made to FIG. 11. A dielectric layer **180** is formed in the hole **H**. The dielectric layer **180** conformally lines the sidewall **154a** of the second ILD layer **154** and the top surface of the source/drain contact **172**. The dielectric layer **180** includes a bottom horizontal portion **182**, a hole liner **184**, and a top horizontal portion **186** monolithically connected to each other. The bottom horizontal portion **182** is in contact with the top surface of the source/drain contact **172**, and the top horizontal portion **186** is in contact with a top surface of the second ILD layer **154**. The hole liner **184** is in contact with the sidewall **154a** of the second ILD layer **154** in the hole **H**. The hole liner **184** can be referred to as a dielectric liner in some embodiments. In some embodiments, the dielectric layer **180** may be formed in the hole **H** by a suitable technique, such as chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD) other suitable processes, or combinations thereof. In some embodiments, the dielectric layer **180** may include dielectric materials such as SiN, SiC, SiCN, SiON, AlO, AlON, ZrO, ZrN, the like or combinations thereof. A thickness of the dielectric layer **180** may range from about 1 nm to about 5 nm. In some embodiments, the portions of the gate spacers **130** exposed by the hole **H** are in contact with the dielectric layer **180**.

A thickness of the dielectric layer **180** may be in a range from 1 nanometer to 5 nanometers. In some embodiments, the dielectric layer **180** may be formed under a pressure in a range of 1 mTorr to 10 mTorr and in a suitable temperature range. For example, in some embodiments, a SiN layer is formed in the temperature range from about 250° C. to about 500° C. with a precursor of dichlorosilane (DCS)/NH<sub>3</sub>. In some embodiments, a SiC layer is formed in the temperature range from about 200° C. to about 450° C. with a Si—C contained precursor, such as tetramethylsilane (TMS). In some embodiments, a SiON layer is formed in the temperature range from about 200° C. to about 450° C. with a precursor of SiH<sub>4</sub>/N<sub>2</sub>O. In some embodiments, a SiCN layer is formed in the temperature range from about 200° C. to about 450° C. with a precursor of Si—C contained reaction gas. In some embodiments, a AlO/AlON layer is formed in the temperature range from about 200° C. to about 400° C. with a precursor of trimethylaluminum (TMA)/H<sub>2</sub>O. In some embodiments, a ZrO/ZrN layer is formed in the temperature range from about 200° C. to about 400° C. with a precursor of ZrCl<sub>4</sub>/H<sub>2</sub>O.

Reference is made to FIG. 12. The bottom horizontal portion **182** and the top horizontal portion **186** of the dielectric layer **180** (as shown in FIG. 11) are removed by an etching process, such as a dry etching, a wet etching, and/or other etching methods (e.g., reactive ion etching). The hole liner **184** of the dielectric layer **180** remains on the sidewall **154a** of the second ILD layer **154** and over the source/drain contact **172**. The remaining hole liner **184** has a bottom surface in contact with the source/drain contact **172**, and

hence the bottom surface of the hole liner **184** is in a position lower than top surfaces of the gate electrodes **160**. The remaining hole liner **184** defines an opening **1840** to expose a portion of the source/drain contact **172**. Bottom and top surfaces of the source/drain contact **172** are respectively in contact with the epitaxy feature **140** and the hole liner **184**. In some embodiments, an anisotropic etching process may be performed. In some embodiments, the etching process includes reacting an etchant with the dielectric layer **180**, in which the etchant has high selectivity between the second ILD layer **154** and the dielectric layer **180**. In other words, the second ILD layer **154** and the dielectric layer **180** have different etch resistance properties. For example, the second ILD layer **154** has higher etch resistance to the etchant used to etch the dielectric layer **180** than that of the dielectric layer **180**.

Next, a conductive structure **174** is formed in the contact hole H and in contact with the source/drain contact **172**, and a CMP process is carried out to planarize the conductive structure with top surfaces of the second ILD layer **154** and of the dielectric layer **180**. The resulting structure is shown in FIG. **13**. To be specific, the conductive structure **174** is formed in the opening **1840** defined by the hole liner **184**, such that the hole liner **184** surrounds and is in contact with the conductive structure **174**. The hole liner **184** thus separates the conductive structure **174** from the second ILD layer **154**. In the cross-sectional illustration, there are hole liners **184** respectively in contact with opposed sidewalls of the hole H and arranged on the top surface of the source/drain contact **172** in a spaced-apart manner, and the conductive structure **174** is formed between the hole liners **184**.

In illustrated embodiments, a bottom surface of the hole liner **184** is lower than top surfaces of the gate electrodes **160**, and a top end of the hole liner **184** is substantially level with a top surface of the conductive structure **174**. Because of the hole liner **184**, a width of the conductive structure **174** is less than a width of the source/drain contact **172**. Exemplary formation method of the conductive structure **174** may include depositing metal or other suitable conductive materials in the source/drain contact hole O1 using a deposition process, such as a CVD process.

Reference is made to FIG. **14**. An etch stop layer **190** is formed over the second ILD layer **154** and the conductive structure **174**. The etch stop layer **190** may include a dielectric material, such as SiN, SiC, SiCN, SiON, the like, or combinations thereof. In some embodiments, the etch stop layer **190** may include material the same as that of the hole liner **184** and/or that of the gate spacers **130**. In some embodiments, the etch stop layer **190** may be deposited using chemical vapor deposition (CVD), high density plasma (HDP) CVD, sub-atmospheric CVD (SACVD), molecular layer deposition (MLD), sputtering, physical vapor deposition (PVD), plating, or other suitable techniques.

In some embodiments, a thickness of the etch stop layer **190** is greater than that of the hole liner **184**. For example, the thickness of the etch stop layer **190** is in a range of 5 nanometers to 50 nanometers. In some embodiments, the etch stop layer **190** is a SiN layer formed in the temperature range from about 250° C. to about 500° C. with a precursor of DCS/NH<sub>3</sub>. In some embodiments, the etch stop layer **190** is a SiC layer formed in the temperature range from about 200° C. to about 450° C. with a precursor containing Si—C. In some embodiments, the etch stop layer is a SiON layer formed in the temperature range from about 200° C. to about 450° C. with a precursor of SiH<sub>4</sub>/N<sub>2</sub>O. In some embodiments, the etch stop layer **190** is a SiCN layer formed in the

temperature range from about 200° C. to about 450° C. with a precursor containing Si—C or N reaction gas.

Then, a third ILD layer **156** is formed over the etch stop layer **190**. In some embodiments, the third ILD layer **156** may include material the same as that of the second ILD layer **154** and/or that of the first dielectric layer **152** (as shown in FIG. **7**). The third ILD layer **156** may be composed of any suitable dielectric or insulating material such as, but not limited to, silicon dioxide, SiOF, carbon-doped oxide, a glass or polymer material. For example, the dielectric material of the third ILD layer **156** may include tetrathoxysilane (TEOS), an extreme low-k (ELK) dielectric material, nitrogen-free anti-reflective coating (NFARC), silicon oxide, silicon nitride, silicon oxynitride, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), spin-on glass (SOG), fluorinated silica glass (FSG), carbon doped silicon oxide (e.g., SiCOH), Xerogel, Aerogel, amorphous fluorinated carbon, Parylene, BCB (bis-benzocyclobutenes), polyimide, the like, or combinations thereof. The ELK dielectric material has a dielectric constant less than, for example, about 2.5. It is understood that the third ILD layer **156** may include one or more dielectric materials and/or one or more dielectric layers. In some embodiments, the third ILD layer **156** may be deposited by chemical vapor deposition (CVD), high density plasma (HDP) CVD, sub-atmospheric CVD (SACVD), spin-on coating, sputtering, or other suitable techniques. In some other embodiments, the third ILD layer **156** may include multiple layers of the same or differing dielectric materials may instead be used.

Reference is made to FIG. **15**. A gate contact hole O2 is formed in the third ILD layer **156**, the etch stop layer **190** and the second ILD layer **154** until reaching one or more gate electrodes **160**, such that the one or more gate electrodes **160** are exposed. The gate contact hole O2 may be formed by one or more etching processes. The one or more etching processes stops at the top surface of the gate electrode **160**. Since the top surface of the gate electrode **160** is in a position higher than a bottom of the hole liner **184**, a bottom of the gate contact hole O2 may be in a position higher than the bottom of the hole liner **184** as illustrated. For example, a first opening **1560** is etched in the third ILD layer **156** using a first etching process. In some embodiments, the etch stop layer **190** may have higher etch resistance to the first etching process than that of the third ILD layer **156**. A second opening **1900** is etched in the etch stop layer **190** through the first opening **1560** using a second etching process. A third opening **1540** is etched in the second ILD layer **154** through the first opening **1560** and the second opening **1900** using a third etching process. The first to third openings **1560**, **1900**, and **1540** are spatially communicated to form the gate contact hole O2. Etchant used in the first to third etching processes may be different. For example, the first and third etching process may use an etchant such as dilute hydrofluoric acid (HF), HF vapor, CF<sub>4</sub>, C<sub>4</sub>F<sub>8</sub>, CH<sub>3</sub>F<sub>3</sub>, C<sub>2</sub>F<sub>6</sub>, SF<sub>6</sub>, or NF<sub>3</sub> gas, and the second etching process may use an etchant such as CH<sub>2</sub>F<sub>2</sub> or other applicable etchants. In some embodiments, the hole liner **184** has higher etch resistance to the first etching process than that of the third ILD layer **156**, and the conductive structure **174** is thus protected from the first etching process by hole liner **184**. In some embodiments, the hole liner **184** has higher etch resistance to the second etching process than that of the etch stop layer **190**, and the conductive structure **174** is thus protected from the second etching process by hole liner **184**. In some embodiments, the hole liner **184** has higher etch resistance to the third etching process than that of the second ILD layer **154**, and the conductive structure

174 is thus protected from the third etching process by the hole liner 184. Since the hole liner 184 protects the conductive structure 174 from one or more etch processes for forming the gate contact hole O2, process window of lithography and/or etch processes for forming the gate contact hole O2 can be relaxed. Therefore, an enlarged gate contact hole O2 can be formed over the gate G.

Reference is made to FIG. 16. A gate contact 200 is formed in the gate contact hole O2. In illustration, the hole liner 184 is between the conductive structure 174 and the gate contact 200, and the ILD layer 154 is between the hole liner 184 and the gate contact 200. The gate contact 200 is a conductor, such as aluminum, aluminum copper or copper. For example, a conductive layer may be deposited over the ILD layer 156 and filling the gate contact hole O2, an excess portion of the conductive layer outside the gate contact hole O2 is removed, using for example, a CMP process, and thereby forming the gate contact 200. The conductive layer may be formed, for example, by PVD, CVD, ALD, electroplating (ECP) step, a chemical electroless deposition (ELD), or various combinations thereof. As aforementioned, an enlarged gate contact hole O2 can be provided because the conductive structure 174 is protected from the etching the gate contact hole O2, and hence an enlarged gate contact 200 can be formed in such an enlarged gate contact hole O2, which is advantageous for reduction of contact resistance between the gate contact 200 and the gate electrode 160.

Based on the above discussions, it can be seen that the present disclosure offers advantages over interconnection structures. It is understood, however, that other embodiments may offer additional advantages, and not all advantages are necessarily disclosed herein, and that no particular advantage is required for all embodiments. One advantage is that an undesirable electrical connection between the gate contact and the source/drain contact can be prevented because the hole liner protects the conductive structure over the source/drain contact against etching the gate contact hole. Another advantage is that the process window for forming gate contact holes can be relaxed. Yet another advantage is that the gate contact hole and the gate contact can be enlarged and thus reduce the gate contact resistance.

According to some embodiments of the present disclosure, a semiconductor device includes a semiconductor substrate, a gate electrode, a source/drain contact, a conductive structure, and a dielectric liner. The semiconductor substrate has a channel region and a source/drain region. The gate electrode is over the channel region. The source/drain contact is over the source/drain region. The conductive structure is over a top surface of the source/drain contact. The dielectric liner surrounds the conductive structure and is over the top surface of the source/drain contact.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the

spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor device, comprising:

a semiconductor substrate having a channel region and a source/drain region;

a gate structure over the channel region;

a gate spacer formed on a sidewall surface of the gate structure;

a source/drain contact over the source/drain region;

an interlayer dielectric layer formed over the substrate;

a conductive structure formed in the interlayer dielectric layer and over a top surface of the source/drain contact;

a dielectric liner disposed between the conductive structure and the interlayer dielectric layer and separating sidewall surfaces of the conductive structure from the interlayer dielectric layer, wherein a bottom surface of the dielectric liner is below a top surface of the gate spacer;

a gate contact structure extending upwardly from the gate structure;

a second dielectric layer over the interlayer dielectric layer, wherein a top surface of the dielectric liner is below a bottom surface of the second dielectric layer; and

wherein the gate contact structure extends through the second dielectric layer.

2. The semiconductor device of claim 1 wherein the structure comprises a gate electrode, and wherein the dielectric liner contacts the gate spacer.

3. The semiconductor device of claim 1 wherein the gate contact structure is electrically isolated from the conductive structure by the dielectric liner and the interlayer dielectric layer.

4. The semiconductor device of claim 1 wherein an interface between the dielectric liner and the source/drain contact is lower than the top surface of the gate spacer.

5. The semiconductor device of claim 1 wherein top surfaces of the conductive structure, the interlayer dielectric layer and the conductive structure are co-planar.

6. The semiconductor device of claim 1 wherein the source/drain region comprises one or more epitaxial layers.

7. The semiconductor device of claim 6 wherein the one or more epitaxial layers comprise silicon.

8. The semiconductor device of claim 6 wherein the one or more epitaxial layers comprise silicon germanium.

9. The semiconductor device of claim 1 wherein the substrate comprises a fin and wherein the source/drain region is formed in the fin.

10. The semiconductor device of claim 1 wherein the dielectric liner contacts the gate spacer.

11. The semiconductor device of claim 1 wherein a bottom end of the dielectric liner and a bottom surface of the conductive structure are in contact with the source/drain contact.

12. The semiconductor device of claim 1 wherein the gate contact structure is formed by two or more processes.

\* \* \* \* \*